

# M15 - Azure + Microsoft Fabric Data Engineering

Beginner-First Professional Institute Edition

**OPTION B | RECOVERED SUCCESSOR SEAL**

**Historical lost seal is not being impersonated.**

# M15 Start Here - What this module does

## This book is your teacher

M15 is the first dedicated cloud/Fabric module. New platform words are introduced before use. Visuals show the interface; the course controls the exact practice, validation, evidence and repair route.

## Module outcome

Design and, where access permits, execute defensible Microsoft Fabric data-engineering work using OneLake, Lakehouse/Delta, notebooks/Spark, pipelines, SQL serving, security, monitoring and lifecycle controls.

## Truth rule

LIVE FABRIC means you actually observed the named item/path/run/permission/data state. OFFLINE PLAN is design only. SIMULATOR is deterministic local evidence. Never convert one category into another by confidence.

# Exact study loop

## Normal lesson route

- Open Teaching Book at mapped page.
- Use current source/visual only for the named concept.
- Follow the guided page once.
- Close source and complete independent page.
- Save genuine attempt/evidence.
- Only then open protected Review if repair is needed.
- Close Review and complete changed retry.
- Teach back and move only with evidence.

## Failure route

saved attempt -> targeted teaching repair -> changed micro-proof -> validation -> reassessment. Failed attempts remain append-only.

# First-use tool and access map

## Browser + authorized Fabric access

Use your real authorized tenant/trial/capacity if available. Do not invent tenant screenshots, capacity state or run IDs.

## Local simulator

Python is used only for deterministic controls on bundled fixtures. Passing simulator checks is NOT live Fabric execution.

## Runtime authority

(Spark 3.5.5 / Python 3.11 / Java 11 / Delta 3.2). Runtime 2.0 is GA (opt-in)  
Public Preview (Spark 4.1 / Python 3.13 / Java 21 / Delta 4.2).

# Module map and exact routes

## Lessons 01-05

Cloud mental model -> identity -> OneLake -> workspace/items -> paths/shortcuts.

## Lessons 06-10

Lakehouse/Delta -> notebooks -> Fabric runtime -> pipelines -> SQL endpoint/Warehouse choice.

## Lessons 11-16

Medallion -> incremental MERGE -> security/connections -> monitoring/cost -> lifecycle -> end-to-end case.

## Assessment route

M15 lesson mastery -> Module Practical -> M16. G06 remains locked until M15 + M16 + M17 evidence is complete.

# CREATE vs FILL and evidence policy

## CREATE

Your attempt.md, retry.md, evidence notes/screenshots/exports and live run proof are learner-created. They are intentionally absent from the package until you create them.

## FILL

Use supplied blank templates/fixtures where present; never overwrite raw source or protected Review.

## Protected Review

Review is diagnosis after a saved attempt, not a pre-solved answer source.

## No percentage shortcut

Every required competency needs evidence and zero unresolved critical failures. An average cannot hide missing live/runtime proof.